Section A

Answer all questions in the space provided

Question 1

Restriction enzymes cut DNA molecules only at specific sites with particular base sequences. The target sites for the enzyme EcoR1, Bam H1 and Hpa 1 are shown in Table 1.1 below. The lines drawn in each sequence show where the enzyme cuts the DNA molecule.

restriction enzyme	specific target base sequence of DNA
EcoRI	G A A T T C C T T A A G
BamHI	G G A T C C C C T A G G
HpaI	G T T A A C C A A T T G

Table 1.1

(a) State the type of ends created when cutting DNA with the restriction enzyme.

(i) EcoR1**: sticky end [1]**

(ii) Hpa 1: **Blunt end [1]**

(b) The target base sequences for all three restriction enzymes in Table 1.1 are palindromic.

(i) With reference to Table 1.1, explain what is meant by the term palindromic. [1]

-DNA sequence on complementary strand reads the same in opposite directions. [1]

(ii) Suggest why it is preferable that restriction enzyme target sequences be palindromic. [1]

-To allow restriction enzymes to cut both strands at the same time to produce fragments. [1]

A bacterial plasmid includes the base sequence shown in Figure 1.2 below. There are no target sequences for the three restriction enzymes in the rest of the plasmid.

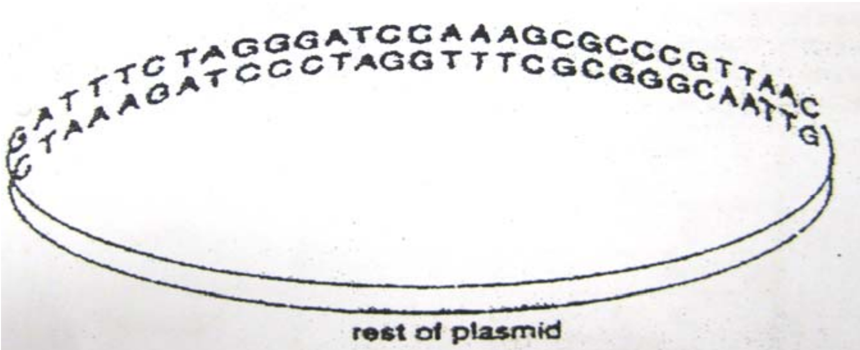

Figure 1.2

(c) State how many fragments of DNA will be present after the plasmid has been treated with:

(i) EcoR1: **0 [1]**

(ii) a mixture of EcoR1 and BamH1**: 1 [1]**

A genetic engineer would like to use the above plasmid in genetic engineering.

(d)(i) Explain why he needs to use a plasmid for the process of genetic engineering.[1]

-As a vector [1/2]
-To carry the DNA of interest into host cell (bacteria). [1/2]

(ii) Suggest which enzyme from Table 1.1 should he use and explain why this enzyme would be the scientist's choice. [3]

- BamH1 [1]
-Sticky ends generated, H bonds formed between complementary base pairs[1]
-no need to add linker DNA [1]
-EcoR1 does not make any cuts on this plasmid [1]
- DNA molecules held in place for DNA ligase to catalyse formation of phosphodiester bonds [1]

Any 3

Question 2

Figure 2.1 shows the three stages involved in the polymerase chain reaction (PCR).

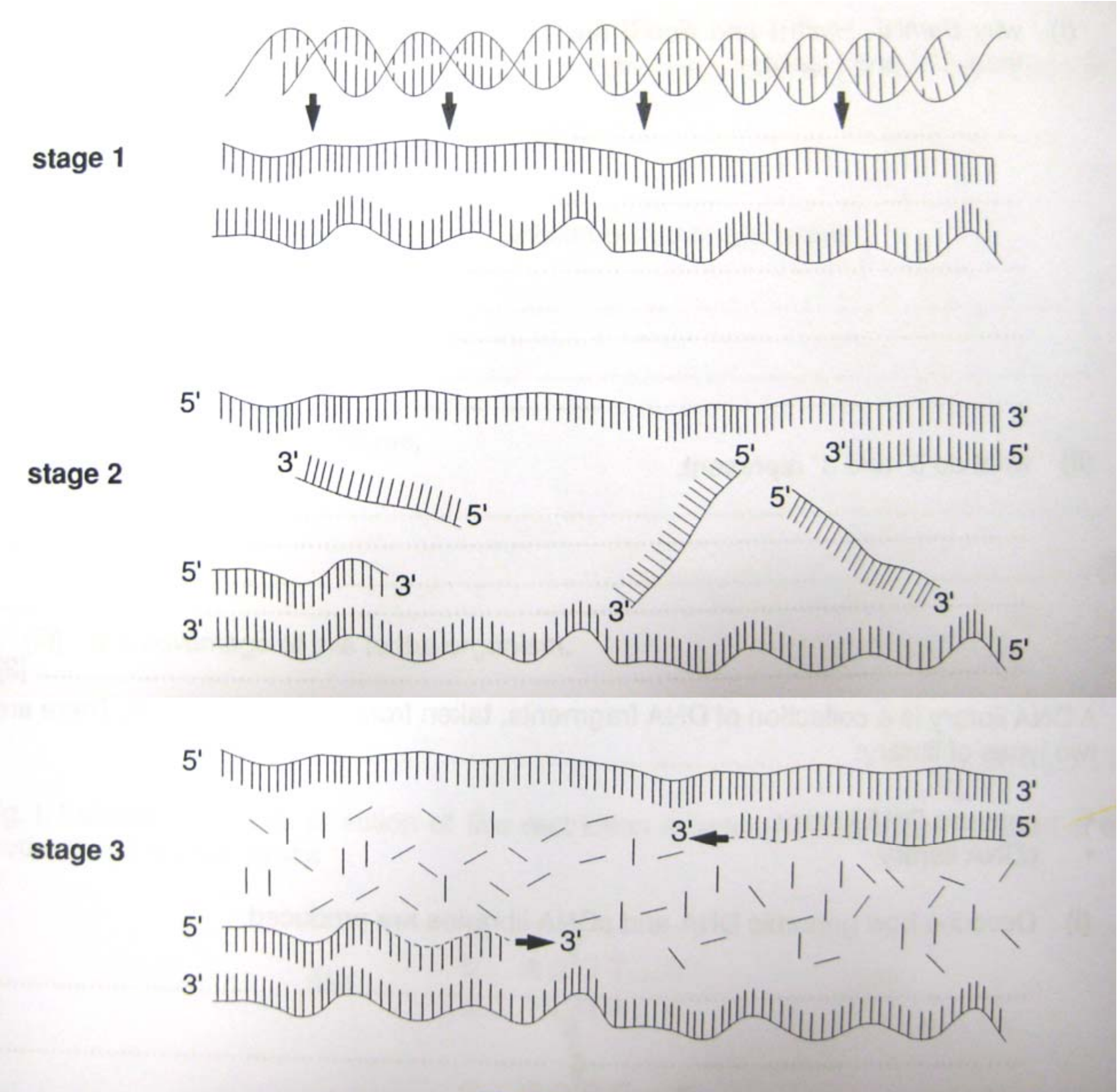

Fig. 2.1

With reference to Figure 2.1, explain what is happening at each of the three stages.
[1/2 mark for each point]

(a) stage 1 [2]

-Heat at 95^{o}C
-H bonds between complementary base pairsbroken
- DNA double helix
-separates into 2 single strands

stage 2 [2]

-Cool to 55oC
-due to an excess of primers added
-primers bind to the complementary sequence on DNA
-instead of DNA reannealing into double strand

stage 3 [2]

-Taq polymerase synthesise daughter strands
-By adding on to the 3' end of the primer
-Deoxyribonuclotides are required as raw material
-Amplification of 2 strands to 4 strands

(b) Why is it necessary to join primer molecules to the single stranded DNA before DNA polymerase is used? [1]

-DNA polymerase can only add on to the 3' end of an existing strand of DNA [1]

(c) State 3 ways in which PCR differs from DNA replication in living cells. [3]

Any 3 of the following
½ mark each

PCR	DNA replication
DNA primer	RNA primer
Does not require DNA ligase/ primase	Require DNA ligase/ primase
Various temperature required	Constant temperature: body temperature
No synthesis of Okazaki fragments	Okazaki fragment synthesis

Question 3

(a) The family with the autosomal dominant nail-patella syndrome required a second evaluation because individual 3 from generation II could not have a prenatal diagnosis using new restriction fragment length polymorphisms (RFLPs) based on her husband's genotype. **Fig. 1.1** illustrates a polymerase chain reaction (PCR) analysis that examines a variable number of tandem repeats (VNTRs) in a DNA region very near the nail-patella locus. Each repeat contains two nucleotides, and the number of repeats varies to produce at least six differently sized alleles when the region is amplified by PCR, using primers P1 and P2.

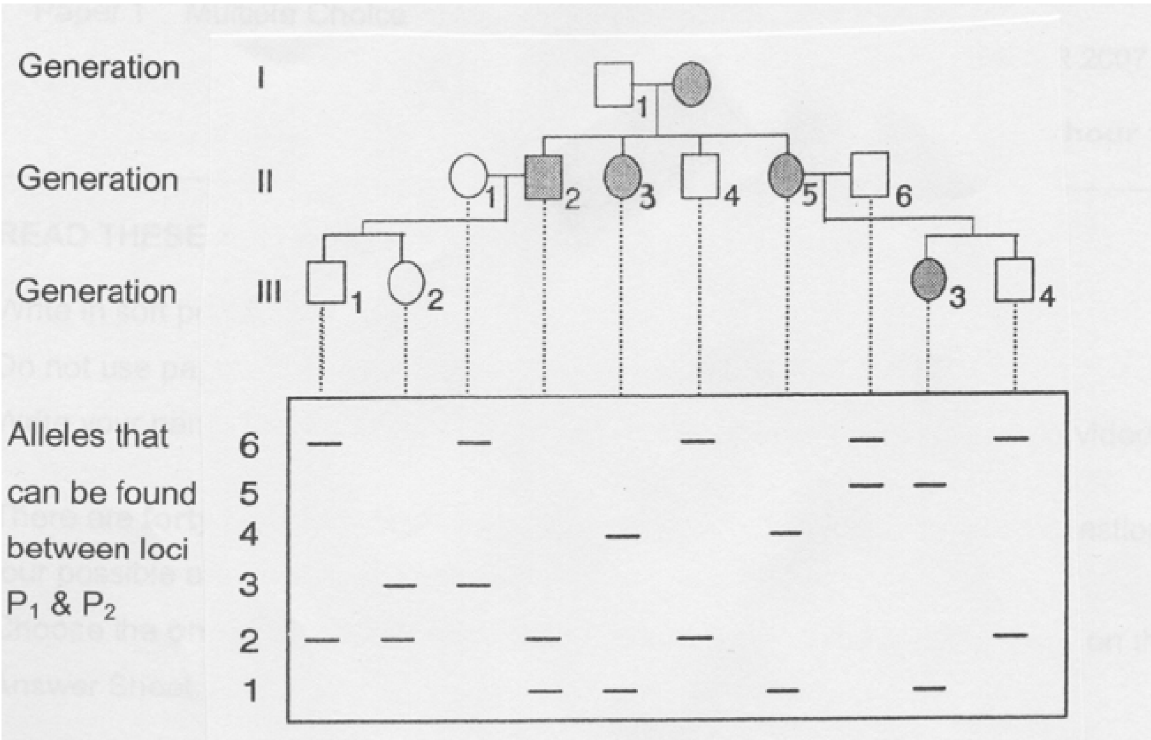

Fig. 1.1

(i) Explain the molecular basis for a restriction fragment length polymorphism. [2]

[1 mark each, max 2]

- **genetic difference among individuals**
- **different restriction fragment patterns produced / variety of lengths of restriction fragments (length *polymorphism*)**
- **digestion of the DNA samples with specific restriction endonucleases.**
- **used as molecular markers**
- **segment of DNA found at a specific site in the genome**

(ii) Based on the family analysis shown in the figure, match the individual 3 from generation II with correct genotype. [1]

- **1, 4 [award mark only if both are correctly stated]**

[Turn over

(iii) Outline how an RFLP can be detected experimentally. [3]

[1 mark each, max 3]

- **Reference to restriction digestion**
- **Reference to agarose gel electrophoresis**
- **(no marks for Southern blotting without elaboration) Hybridization of the membrane to a labeled DNA probe**
- **then determines the size of the fragments which are complementary to the probe**
- **RFLP occurs when the size of a detected fragment varies between individuals.**

(b) Species A contain two different RFLPs which are linked.
RFLP 1 is found in 4500 bp and 5200 bp lengths and RFLP 2 can be 2100 bp or 3200 bp. A homozygote, 4500 bp and 2100 bp, is crossed to a homozygote, 5200 bp and 3200 bp. The F_1 offspring are then crossed to a homozygote containing the 4500 bp and 2100 bp fragments. The following results are obtained:

5200, 4500 and 2100	40 offspring
5200, 4500, 3200 and 2100	98 offspring
4500 and 2100	97 offspring
4500, 2100 and 3200	37 offspring

(i) State the genotypes of the recombinant offspring. [1]

1. 5200, 4500 and 2100 [1/2]

2. 4500, 2100 and 3200 [1/2]

(ii) Calculate the map distance between the RFLPs. [1]

Map distance = (40+37)/ (40 + 98 +97 +37) x 100 [working – ½ mark]
= 28.3 mu [ans – ½ mark (must include units)]

(iii) Explain how RFLP analysis facilitated the process of genomic mapping. [2]
[Max 2]

- **Genomic mapping determines the complete DNA sequence of an organism [1/2]**
- **RFLP analysis facilitates linkage mapping [1/2]**
- **RFLP as molecular marker which is analogous to an allele of a gene [1]**
- **Linkage mapping uses the outcome of genetic crosses to map the relative locations of genes along chromosomes [1]**
- **Determine locations of functional genes by correlating the inheritance of the gene with the inheritance of a particular RFLP. [1]**

Question 4

The following **Figure 4.1** illustrates how stems cells are cultivated for research.

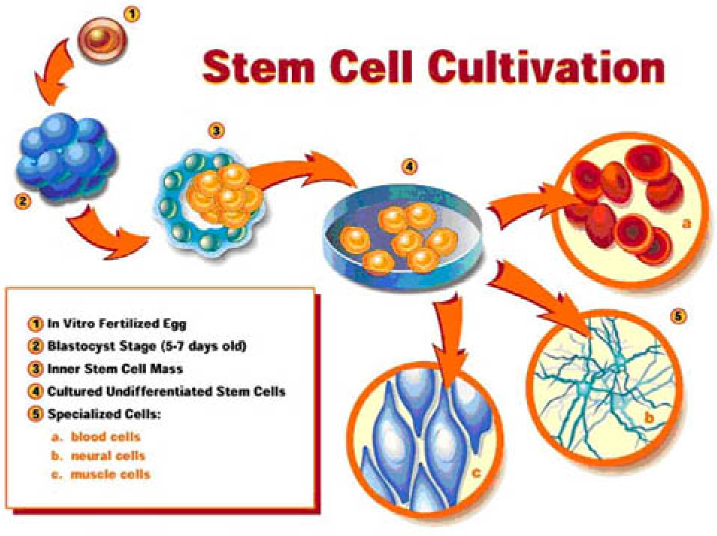

Figure 4.1

(a) Explain what makes a stem cell unique from a normal human adult cell. [3]

- **Capable of self-renewal**
- **Unspecialised**
- **Can differentiate into different cell types**

(b) Name the chemical required to convert the undifferentiated stem cells in Step 4 of Figure 1.1 into the specialised cells in Step 5.[1]

- **Cytokines**

(c) The stem cells cultivated above are from the inner cell mass of the blastocyst. Are these the most useful stem cells available to science? Explain your answer. [3]

- **No**
- **Blastocyst embryonic stem cells are pluripotent**
- **Zygotic embryonic stem cells are totipotent and are therefore more useful**

OR

- **Yes**
- **Although they are pluripotent, they are able to produce all the cell types required to make any human tissue**

[Turn over

(d) State and explain one similarity between a stem cell and a cancer cell.[2]

- **Both are immortal/capable of indefinite self-renewal**
- **Because they both produce telomerase**

(e) (i) Discuss the two ways in which SCID can come about in an individual.[4]

- **Due mutation in the interleukin 2 receptor gamma (IL2RG) gene, on the X-chromosome**
- **Results in lack of communication between immune system cells**
- **Can also be due to mutation to the gene for Adenosine Deaminase(ADA), the enzyme adenosine deaminase is not produced**
- **Leads to increased levels of dATP which are toxic to immune system cells**

(e) (ii) Suggest how stem cells could be used to enhance gene therapy for patients suffering from Severe Combined Immunodeficiency (SCID). [2]

- **The normal copy of the gene could be inserted into a stem cell rather than the actual defective cell**
- **The modified stem cell can then produce working copies of the cell on a permanent basis**

[Total: 15 marks]

Question 5

(a) Outline the procedure and discuss the underlying principle for the process of Southern blotting. [15]

1m for each point

Gel electrophoresis

- **Southern blotting involves process of gel electrophoresis, transfer of DNA onto nitrocellulose paper, followed by probing.**
- **Gel electrophoresis is a technique in which charged molecules, such as protein or DNA, are separated according to physical properties such as size or charge as they are forced through a gel by an electrical current.**
- **Agarose gel electrophoresis is used to separate DNA.**
- **The meshwork of polymer fibres that makes up the agarose gel impedes the movement of DNA fragments, affecting the longer fragments more than the shorter ones. Therefore, shorter DNA fragments will move towards the anode at a faster rate than the longer fragments.**
- **DNA which is negatively charged at neutral pH because of its phosphate backbone, is pulled through the agarose gel with an electric field from a negative pole (cathode) towards the positive pole (anode).**
- **Prior to loading, DNA sample is first mixed with a dense loading dye, which helps the DNA sink to the bottom of the wells.**
- **As DNA is invisible to the naked eye, the loading dye also helps us to monitor the progress of electrophoresis.**
- **Usually, *markers* are run in one or two of the lanes. Markers are prepared mixtures of DNA fragments of known sizes, which will form the basis of comparison for the unknown fragments in DNA sample. They are normally loaded into the extreme wells.**
- **The gel, containing samples, is submerged in a special pH-regulated solution, or buffer, containing a nucleic acid-specific dye, ethidium bromide.**
- **This dye produces a strong reddish-yellow fluorescence when exposed to UV radiation. Consequently, after electrophoresis, the nucleic acid can be detected by photographing the gel under UV illumination.**
- **Since DNA sample separates out into fragments of different sizes, based in their different rate of migration, they are seen as discrete bands on the gel slab. Each band consists of DNA molecules of the same length.**

<u>Transfer DNA onto nitrocellulose membrane</u>

- **The gel slab is containing the DNA fragment is placed under a nitrocellulose membrane and a stack of paper towels.**

- **These are placed on top of a sponge in a tray of alkaline solution.**

- **The absorbent paper towels draw the solution towards themselves. This capillary action draws the alkali solution through the gel, denaturing the double stranded DNA fragments.**

- **The single stranded DNA on the gel is drawn upwards to the nitrocellulose membrane.**

<u>Probing</u>

- **Next, a probe consisting of purified, single-stranded DNA corresponding to a specific nucleotide sequence (or mRNA transcribed from that gene) is poured over the nitrocellulose membrane.**

- **After hybridization, the membrane is washed to remove any unhybridized probes.**

- **Any fragment that has a nucleotide sequence complementary to the probe's sequence will hybridize (base-pair) with the probe.**

- **If the probe has been labeled with ^{32}P it will be radioactive, and the sheet will show a band of radioactivity where -the probe hybridized with the complementary fragment when autoradiography is performed.**

- **The process of separating DNA fragments by agarose gel electrophoresis, transferring the fragments onto a membrane filter and hybridizing with a labeled complementary probe is known as Southern Blotting.**

(b) Compare the process of Western and Southern blotting. [5]

Western blot	**Southern blot**
For proteins	**For DNA**
Acrylamide gel	**Agarose gel**
Addition of SDS to coat protein with negative charge	**DNA already negatively charged due to phosphate group in sugar phosphate backbone**
Transfer of protein to nitrocellulose membrane carried out by running an electric current through acrylamide gel (anode) to nitrocellulose paper(cathode).	**Transfer of DNA to nitrocellulose membrane carried out with alkaline buffer by principle of capillary action**
DNA probe used	**Antibodies specific to proteins used`**